

Towards a Crowdsourcing Platform for Low Resource Languages – A Collectivist Approach

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Abstract

text

According to a Facebook study, only 53% of the world’s population have access to encyclopedic knowledge in their first language.¹ Just over half of all tweets are in English (Hong, Convertino, and Chi 2011). The implications of this gap in information by language are substantial, particularly when sources of factual information are limited. For instance the United Nations launched the ‘Verified’ initiative, to combat the adverse effects on public health of false information about COVID-19.³

Machine translation is a challenging problem in itself (Korošec 2011; Okpor 2014), and although there have been some great advances made in recent years, particularly in neural machine translation (Rapp, Sharoff, and Zweigenbaum 2016; Hirschberg and Manning 2015; Lample et al. 2017), only a few systems (Martinus and Abbott 2019; Leventhal et al. 2020; Dossou and Emezue 2020) have been trained to communicate in African languages. In our research, we are working with Bambara, the most widely-spoken language in the Mande family in Western Sub-Saharan Africa that includes 60 to 75 languages spoken by 30 to 40 million people. Bambara, the vehicular language of Mali, has approximately 16 million L1 and L2 speakers. (Vydrin 2018) NMT is currently the most successful approach to generalized machine translation. Used alone, it appears to be unsuited for under-resourced languages such as Bambara due to the lack of the quantity of labeled data (parallel texts) needed. Our hypothesis is that NMT combined with human-in-the-loop crowdsourcing will generate the volume and quality of data needed for successful NMT, improving translation quality and breadth over NMT-

only approaches. In order to obtain a sufficient quantity of labeled data, the project will additionally test the hypothesis that a crowdsourcing interface, coupled with process design, can enable non-professional annotators to contribute data suitable for training an NMT. An overarching hypothesis for this work is that machine translation will enable Bambara speakers to understand information in their native language, and increase literacy, which, in turn will foster social and economic development. This project will not attempt to prove this hypothesis, but it will contribute results that are needed in order to test it.

We are exploring machine translation driven by humans in the loop (González-Rubio, Ortiz-Martínez, and Casacuberta 2012; Carl, Gutermuth, and Hansen-Schirra 2015; Peris, Domingo, and Casacuberta 2017; Alabau et al. 2012) to improve MT models for under-resourced languages through user feedback (Figure 1). In addition to providing needed data, crowdsourcing is a means to engage the Malian public in a technology project capable of furthering the social and economic development of their country and the development of post-colonial identity built around Malian languages and culture (Mohamed, Png, and Isaac 2020; Tomašev et al. 2020; Orife et al. 2020).

To our knowledge, human computation has been applied only sparingly to low-resourced languages. There are good reasons for this, as under-resourced languages bring with them a number of specific challenges, such as non-standard spellings, lack of precise vocabulary for many concepts, lack of a tradition of written, as opposed to oral, expression and the concomitant norms of written expression. Consequentially, there are fewer people with the skills to perform language-based crowdsourcing tasks. It is often the case in the environments where low-resourced languages are used crowdsourcing interfaces may be limited to smartphones or other small-screen platforms, as use of computers and high-bandwidth fixed line internet connections may be rare.

Our work in this direction thus far consists of separate neural machine translation and crowdsourcing modules, based on ... and ... respectively. Figure 2 shows

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³<https://www.un.org/en/coronavirus/\%E2\%80\%98verified\%E2\%80\%99-initiative-aims-flood-digital-space-facts-amid-covid-19-crisis>

	Malian news			Wikipedia		
	Written	Oral	Overall	Written	Oral	Overall
Exact meaning	0.830	0.770	0.840	0.870	0.530	0.730
Literalness	0.730	0.530	0.640	0.830	0.580	0.760
Standard Bambara	0.740	0.790	0.760	0.830	0.850	0.830
Highest BLEU Pair	0.408	0.363	0.408	0.645	0.377	0.645

Table 1: Malian news broadcast and Wikipedia article translation ratings and BLEU scores.

	Malian news			Wikipedia		
	Written	Oral	Overall	Written	Oral	Overall
Exact meaning	0.181	0.208	0.197	0.206	0.186	0.220
Literalness	0.130	0.298	0.243	0.238	0.211	0.244
Standard Bambara	0.234	0.178	0.207	0.171	0.068	0.144

Table 2: Score variance standard deviation

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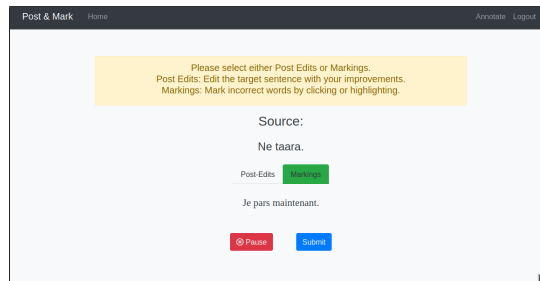


Figure 2: The interface shown when marking the errors in the target language (e.g., French). The interface will be the same while post-editing except that the “Post-Edits” will be highlighted.